

GAIRA V6 — Engine Guide

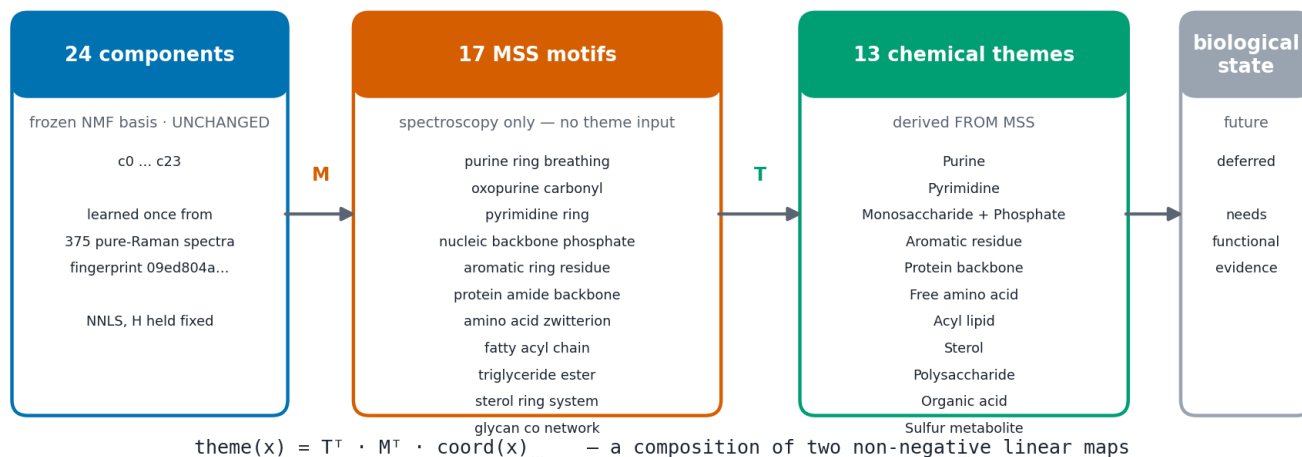
The exact inference pipeline, what every layer does, and how to read the output

GAIRA is not a classifier. It is a frozen coordinate system with two interpretive maps layered on top. Nothing above the atlas is learned: M and T are derived tables, and inference is a projection followed by two matrix products.

1 • The pipeline, in five lines

```
v = preprocess(wavenumber, intensity) # crop → AsLS → SavGol → resample → L2
coord = NNLS(v, H) ; coord /= coord.sum() # H FIXED — the atlas cannot change
mss = MT · coord # 17 spectroscopic motifs
theme = TT · mss # 13 chemical themes
ood = 1 - mean cos to the 5 nearest reference coordinate vectors
```

The V6 semantic hierarchy



V1 fed the component→theme matrix into every MSS weight (25 % of the raw score), so deriving themes from MSS would have been circular. V6 removes that term.

Figure 1 — The layers and what each one means.

2 • Layer by layer

2.1 Preprocessing (frozen)

Crop to 450–1800 cm⁻¹; AsLS baseline ($\lambda=1e5$, $p=0.01$, 8 iterations); Savitzky-Golay (window 9, order 3); resample onto 676 bins at 2 cm⁻¹; L2-normalise. Identical at build time and inference time. Bins outside the input's own range become NaN and are zero-filled at projection — check coverage before trusting a partial spectrum.

2.2 Projection (frozen)

Non-negative least squares onto the 24 × 676 basis with the dictionary held fixed (`update_H=False`). This is **not** cosine matching: the query is fitted as a non-negative combination of the basis spectra. The activations are then L1-normalised, so a coordinate is a *share of the reconstructed evidence* — a

proportion, which is only meaningful because the decomposition is non-negative.

2.3 MSS motifs (V6, new)

A motif is a recurring Raman band pattern, **not a molecule**. M is sparse and non-negative: each column sums to 1 over at most six components. The four evidence lines are band overlap (0.30), basis-spectrum cosine (0.30), exemplar loading (0.30) and perturbation (0.10). No theme label enters any of them.

2.4 Chemical themes (V6, new)

T is a hard partition of the 17 motifs into 13 chemical classes. A theme's score is the sum of its member motifs' activations. Themes are **biochemical classes**, not disease biology — biological-state themes are deliberately not implemented.

3 · How to read the output

output	what it is	how to read it
component coordinates	24 non-negative shares summing to 1	What the atlas actually saw. A sparse, high-contrast vector means a confident fit; a dense flat one means the spectrum is being spread thinly across the basis.
MSS motif composition	17 non-negative scores	What spectroscopy says. In-domain this is the more reliable readout: motif top-1 0.571 / top-3 0.914.
chemical theme composition	13 non-negative scores	The chemistry implied. top-1 0.613 / top-3 0.890 in-domain.
theme confidence	the leading theme's share of total theme mass	Descriptive, not a probability . ECE 0.28 — it over-states accuracy at the top of the range.
expected-theme rank	1–13	The evaluation metric, not an inference output. Reported here so per-theme reliability can be attached to a call.

Per-theme reliability — attach this to any call

chemical theme	n	top-1	top-3	reliability
Purine	9	0.78	1.00	HIGH
Pyrimidine	3	0.67	1.00	MODERATE
Monosaccharide + Phosphate	36	0.72	0.97	HIGH
Aromatic residue	5	0.40	0.80	MODERATE
Protein backbone	23	0.09	0.91	LOW
Free amino acid	11	0.36	0.73	LOW
Acyl lipid	32	1.00	1.00	HIGH
Sterol	12	0.58	0.75	MODERATE
Polysaccharide	5	1.00	1.00	HIGH
Organic acid	8	0.38	0.50	LOW
Sulfur metabolite	9	0.11	0.56	LOW
Flavin + Porphyrin	8	0.88	1.00	HIGH
Carotenoid	2	1.00	1.00	HIGH

Do not report a LOW-reliability theme as a finding. Protein backbone (0.09) and sulfur metabolite (0.11) are ranked first for fewer than one analyte in eight that should express them — even in-domain, on the corpus the atlas was fitted to.

**Part 9 — the radar is now three nested radars, one per abstraction level
component (what the atlas saw) → motif (what spectroscopy says) → theme (what chemistry it implies)**

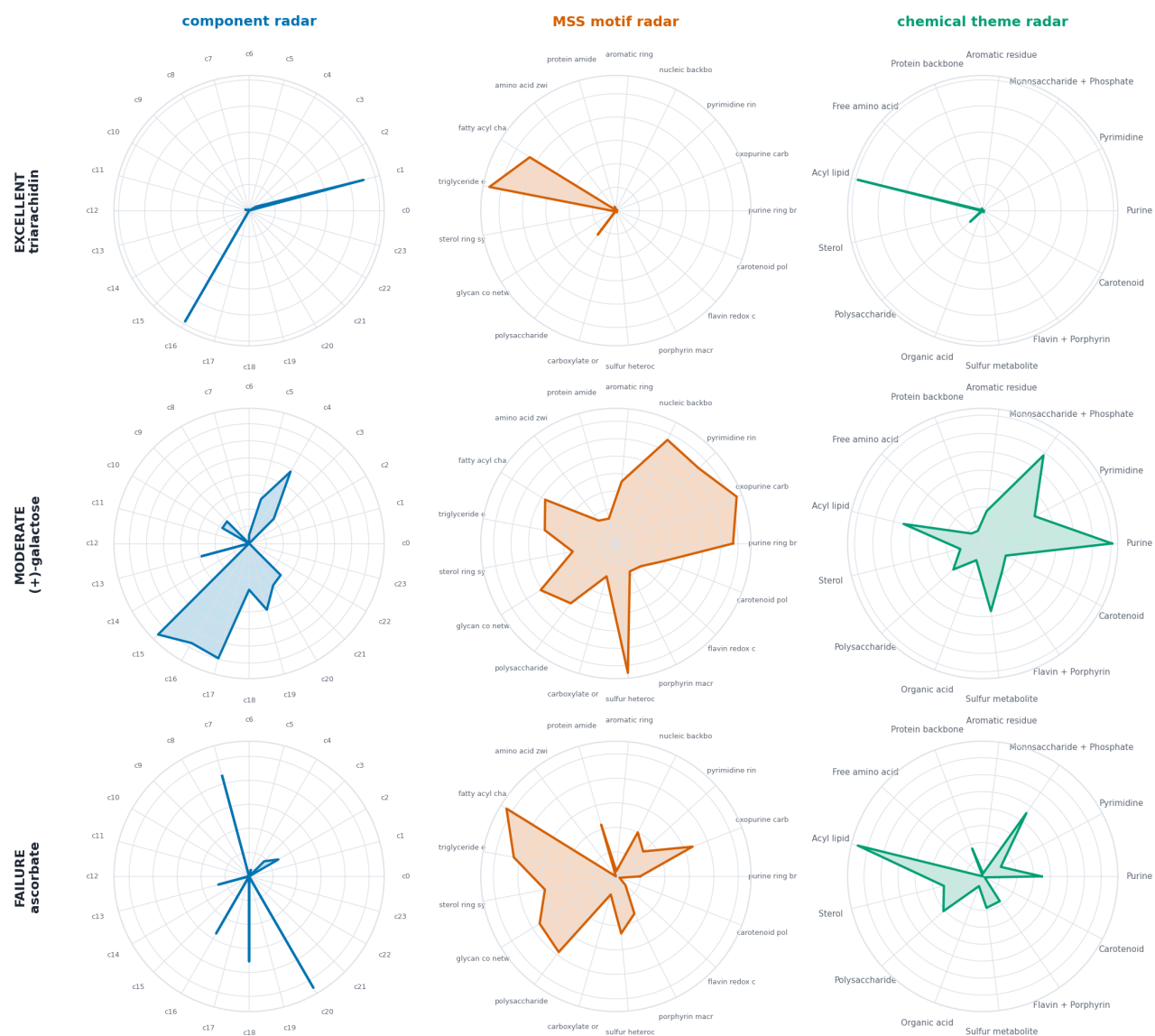


Figure 2 — Read the three radars left to right. A clean component radar with a wrong theme radar is an interpretation failure; a diffuse component radar is a representation failure.

4 · Worked example

Part 10 — one real analyte, end to end, at three performance tiers
green = expected · red = what actually won



Figure 3 — Excellent, moderate and failure cases, end to end.

5 · Scope

- **Validated for:** pure-compound Raman, 450–1800 cm⁻¹, in-domain.
- **Not validated for:** Ag-SERS (theme ranking collapses to chance), serum SERS, Au-SERS, EV, tissue, DART.
- **Never:** molecular identification. GAIRA returns themes and motifs with uncertainty; nearest reference analytes are evidence, never an identification.

Atlas 09ed804a40836f4a05a91ba10900cded. Engine: results/v6_rebuild/code/v6_core/.